

INTEL® UNNATI INDUSTRIAL TRAINING 2025

Project Title: ComicCrafter AI

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Project Number: Problem Statement 6

Introduction

- **ComicCrafterAI aims to democratize comic creation using AI.**

The project seeks to make comic creation accessible to everyone, regardless of their artistic background.

By leveraging advanced AI technologies, ComicCrafterAI provides tools that simplify the creative process, allowing more people to engage in storytelling.

- **Enables storytelling without traditional artistic skills.**

Users can craft compelling narratives and visual stories without needing to draw or design manually.

This opens up opportunities for writers, educators, and hobbyists to express their ideas visually, breaking down barriers to entry in the comic industry.

- **Combines narrative generation with visual creation.**

The system integrates AI-driven text generation with image creation, producing cohesive and engaging comic panels.

This combination ensures that both the story and visuals are aligned, providing a seamless storytelling experience from concept to final product.

Planning and Research

- **Comprehensive research on AI models for local execution.**

The project began with an extensive evaluation of various AI models to ensure they could run efficiently on local machines.

This research was crucial to avoid reliance on cloud infrastructure, making the tool more accessible and cost-effective for users.

- **Selection of Mistral 7B for narrative and Stable Diffusion for images.**

Mistral 7B was chosen for its robust capabilities in generating coherent and engaging narratives suitable for comics.

Stable Diffusion was selected for its ability to create high-quality images, ensuring that the visual output matched the narrative's creativity and style.

- **Focus on output quality and computational efficiency.**

I prioritized models that delivered high-quality results while being computationally efficient, allowing them to run on consumer-grade hardware.

This focus ensured that users could generate comics quickly and effectively without needing high-end computing resources, broadening the tool's accessibility.

- **Challenges in text generation for comic narratives.**

One of the primary challenges was ensuring that the AI could generate text that adhered to the unique structure and style of comic narratives.

This involved understanding complex elements like scene composition, character development, and transitions.

- **Development of a dual-phase generation methodology.**

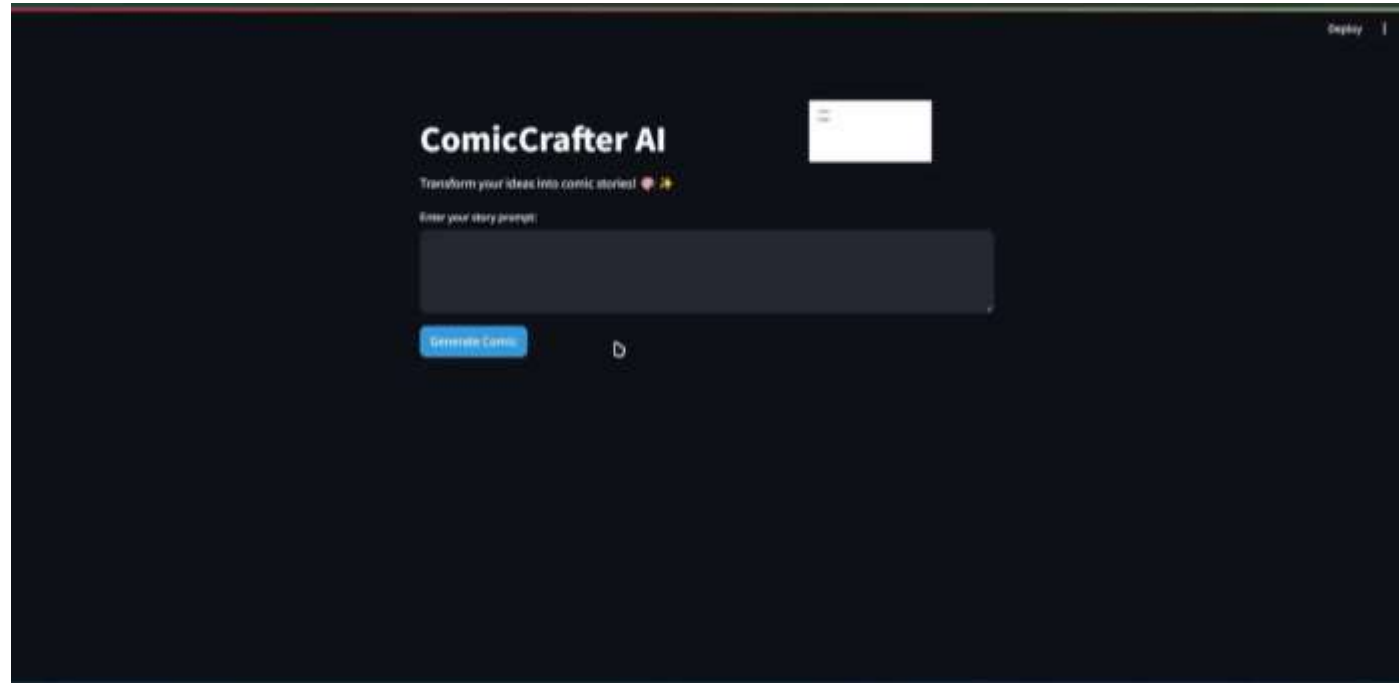
To address these challenges, a dual-phase generation approach was implemented, starting with a broad narrative framework.

This initial phase was followed by an expansion into detailed scene descriptions.

- **Improved narrative consistency through detailed scene descriptions.**

By focusing on detailed scene descriptions, the project achieved greater narrative consistency, ensuring that each panel logically followed the previous one.

Core Development



- **Challenges in maintaining character consistency across panels.**

A significant challenge was ensuring that characters remained visually consistent throughout different panels.

This required sophisticated techniques to manage variations in character positioning, expressions, and interactions while maintaining a recognizable appearance.

- **Use of prompt engineering for visual continuity.**

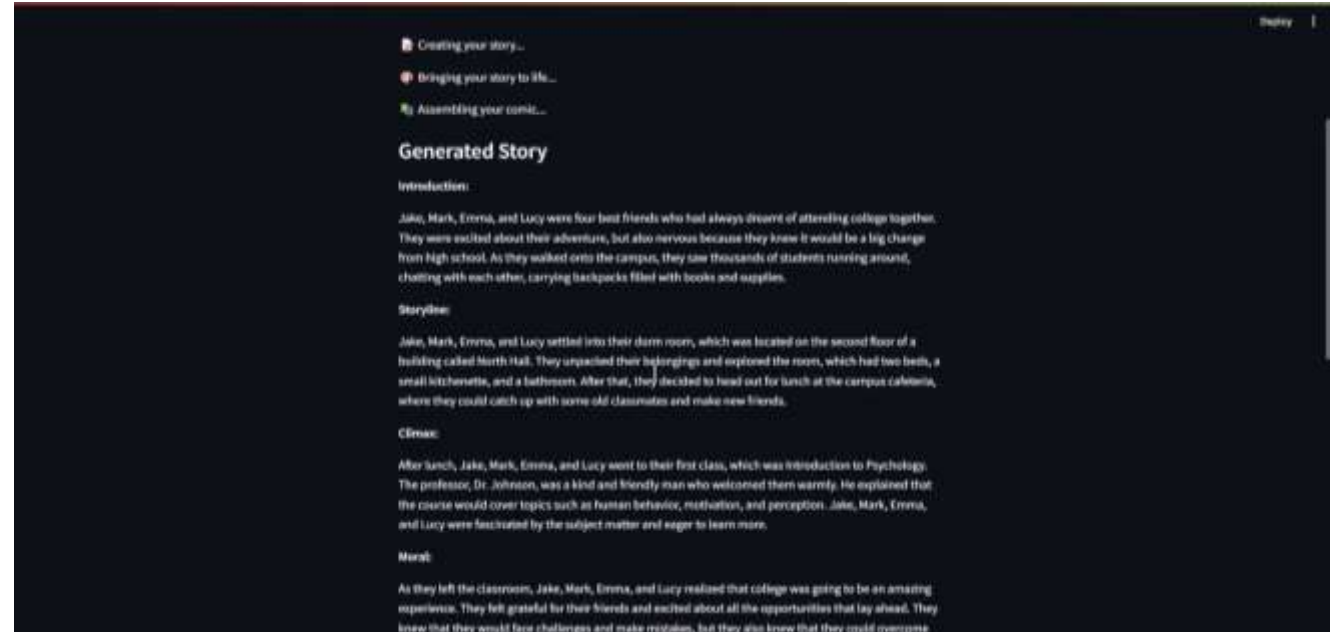
Prompt engineering was employed to guide the AI in generating images that maintained continuity across panels.

By carefully crafting prompts, the system could produce visuals that aligned with the narrative and preserved character and scene consistency.

- **Development of a composition template library for diverse visual presentations.**

To accommodate the diverse visual needs of comic panels, a library of composition templates was developed.

Visual Integration



- **Iterative UI design using Streamlit for balance between simplicity and control.**

The user interface was developed using Streamlit, focusing on creating a balance between ease of use and providing creative control to users.

Through five major iterative revisions, the UI evolved from being overly complex to overly simplified.

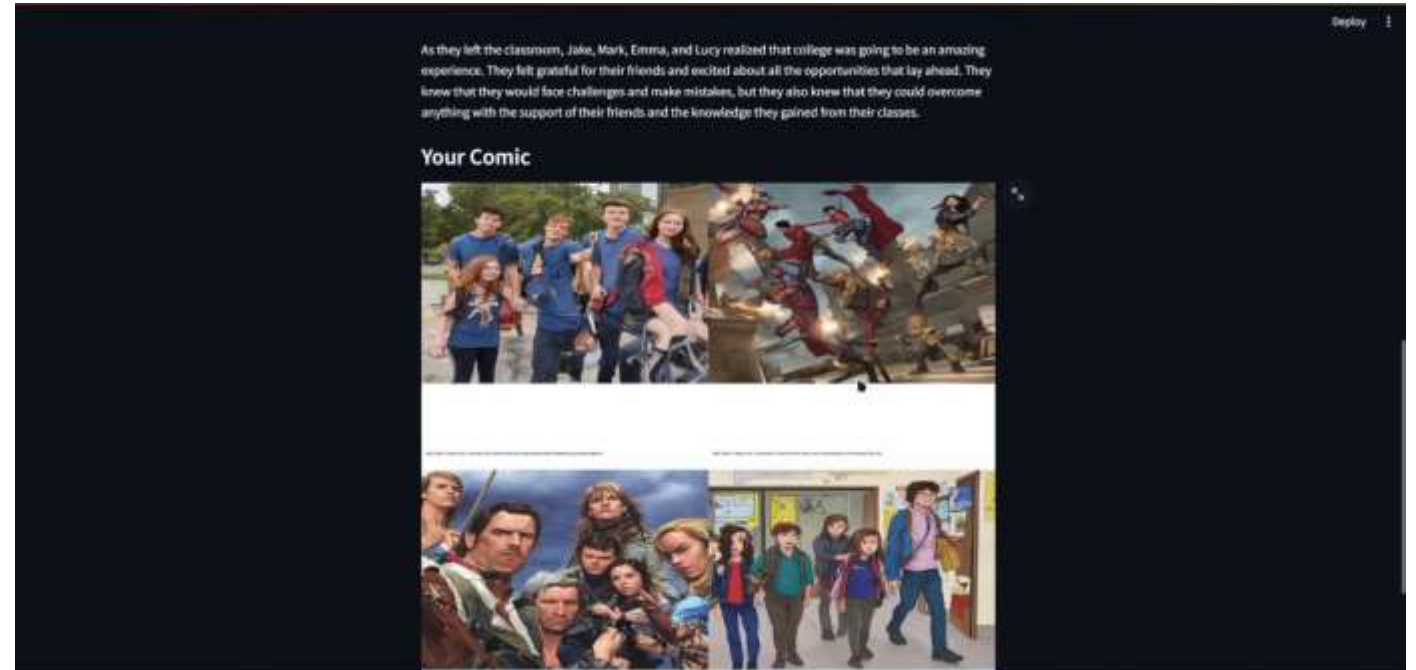
- **Optimization for consumer hardware, reducing resource requirements.**

Significant efforts were made to optimize the system for consumer-grade hardware, ensuring that it could run efficiently without requiring high-end resources.

- **Achieving functionality on 8GB RAM with GPU acceleration.**

The system was fine-tuned to operate effectively on machines with as little as 8GB of RAM, though GPU acceleration remains necessary for optimal performance.

User Interface and Optimization



Conclusion

In concluding the presentation, it's important to reflect on the valuable insights gained from the AI implementation and the development of creative tools.

These insights have highlighted the critical role of prompt engineering and user experience design in creating a viable product.

Looking ahead, the project has set clear future priorities, including expanding the range of artistic styles available, enhancing character consistency across extended narratives, and further optimizing the system for various hardware configurations.

These efforts aim to make the tool even more versatile and accessible.

The project remains in active development, with user feedback playing a crucial role in guiding improvements and ensuring that the tool continues to meet the needs of its users.

This ongoing collaboration with the user community is essential for refining the tool and exploring new possibilities in AI-driven comic creation.